BIT 4131 Statistical Computing and Reporting

CAT 1

Answer all Questions

Acknowledge use of AI/LLM if any and also indicate your own critical thinking. Put references and citations where necessary. APA 6th edition

QUESTION ONE

A university evaluates its student-support programme to improve performance and plan resources. Its anonymised student records contain StudentID, Programme, StudyHours, Attendance and Score.

1. Define a numerical variable and a categorical variable using Score and Programme from the university study. (2 marks)
2. State two measures of dispersion suitable for summarising student scores. (2 marks)
3. State two common file formats for importing the university's student records. (2 marks)
4. State two ethical precautions when reporting student data. (2 marks)
5. Describe why StudyHours and Score are appropriate variables for investigating the relationship between students' study time and examination performance. (3 marks)
6. Explain how duplicate student records could distort the university's score summaries. (3 marks)
7. Distinguish the variance of a sample of student scores from the variance of all scores in the university population. (3 marks)
8. Explain why the student-support report should include assumptions and limitations. (3 marks)
9. Write Python or R code to read the comma-separated values file students.csv and display the number of rows and columns. (3 marks)
10. For scores = [50, 60, 60, 70, 80], write Python or R code to calculate the mean, median and sample variance. (4 marks)
11. Write Python or R code to count missing values in each column of the student data frame. (3 marks)

QUESTION TWO

The university estimates the proportion of high-scoring students and uncertainty in score summaries. For its simulation, examination scores are modelled as normal with mean 65 and standard deviation 10.

1. Define conditional probability and independence in relation to attending support sessions and passing an examination. (2 marks)
2. Explain the difference between the standard deviation of individual scores and the standard error of a sample mean. (3 marks)
3. Write Python or R code to calculate P(Score > 75) (4 marks).
4. The theoretical standard error is 2, but an analyst reports 10 for the simulated means. Analyse the discrepancy and identify two checks on the sampling and summarisation code. (5 marks)
5. Design a bootstrap study to estimate uncertainty in the median of scores = [50, 55, 58, 60, 62, 68, 72, 80]. (6 marks)

QUESTION THREE

Students choose either support method A or B. Independent samples have scores A = [68, 72, 75, 70, 74, 71] and B = [60, 65, 63, 66, 62, 64]. A separate completion study records counts [[40, 10], [30, 20]], with rows A and B and columns completed and not completed. Use a 5% significance level.

1. Explain the null and two-sided alternative hypotheses for comparing the groups' population mean scores. (4 marks)
2. Write Python or R code to perform Welch's independent-samples t-test and display its statistic and p-value (5 marks).
3. For the six students in group A, support-session attendance percentages are [60, 70, 75, 80, 85, 95], paired in order with their supplied scores. Write Python or R code to calculate Pearson correlation (4 marks)
4. Evaluate the claim that a significant difference between groups A and B proves one method caused better performance when students selected their own groups. (3 marks)
5. Satisfaction ratings for three independent student-support groups are ordinal and strongly skewed. Select and justify a suitable non-parametric test (4 marks)

QUESTION FOUR

The university models the relationship between weekly study hours x = [1, 2, 3, 4] and scores y = [50, 55, 65, 70]. For broader student profiles, the data frame students contains StudyHours, Attendance and Score.

1. Explain the roles of the study-hours coefficient and a residual in the university's score model. (2 marks)
2. Write Python or R code to form an intercept-and-predictor design matrix, fit least squares and generate fitted values and residuals (4 marks).
3. For covariance matrix [[1, 0.8], [0.8, 1]], obtain eigenvalues and percentage variance explained by each principal component (4 marks).
4. In a larger regression dataset, residual spread increases with fitted values. Analyse the affected assumption and a consequence for inference, and propose one response. (4 marks)
5. Design a principal component analysis workflow in Python or R for StudyHours, Attendance and Score in students. (3 marks)
6. `The university models passing as log(p / (1 − p)) = −2 + 0.5x, where p is the passing probability and x is weekly study hours. An analyst interprets 0.5 as a 50% increase in passing probability for each extra study hour. Evaluate this claim using the odds ratio and explain the correct interpretation. (3 marks)`

QUESTION FIVE

1. Write a program that calculates the sum of cubes of positive integers from 1 to n for a given value of n, i.e., $\sum_{i}^{n} i^3$ Check your code against the direct formula

$$\sum_{i}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2$$

for different values of n, such as n = 3, n = 30, and n = 300. (5 marks)

2. Identify and describe five errors encountered in numerical computation (5 marks)

3. Briefly describe the features that make a problem solvable by computational methods. (5 marks)

4. Write a code using R or python programming to create a data frame with appropriate headings from the list (5 marks)